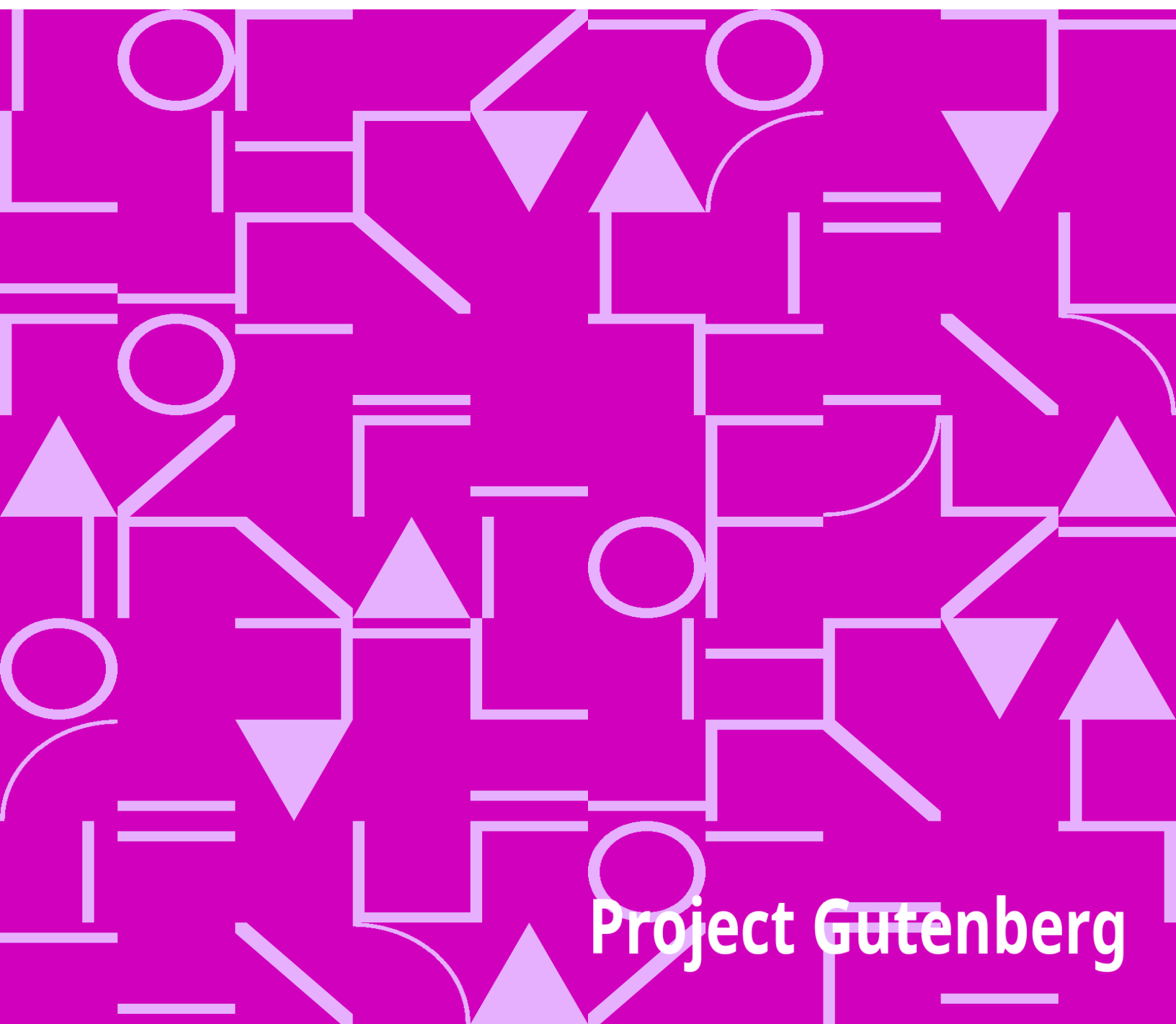


Two New Pelycosaurs from the Lower Permian of Oklahoma

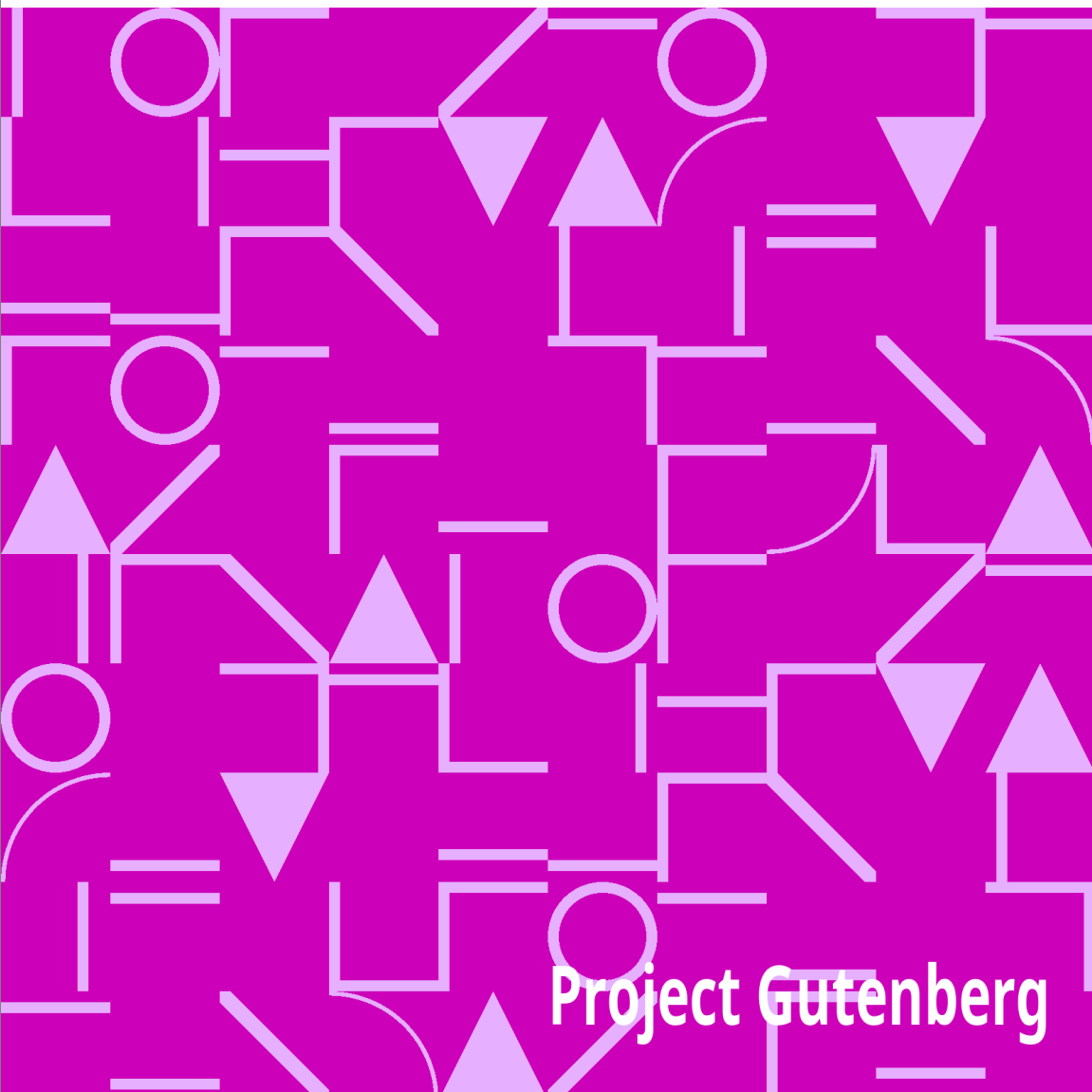
Richard C. Fox



Project Gutenberg

Two New Pelycosaurs from the Lower Permian of Oklahoma

Richard C. Fox

A decorative pattern of various geometric shapes, including circles, triangles, squares, and lines, arranged in a complex, overlapping manner. The pattern is primarily in shades of purple and magenta, with some white outlines.

Project Gutenberg

The Project Gutenberg eBook of Two New Pelycosaurs from the Lower Permian of Oklahoma

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Two New Pelycosaurs from the Lower Permian of Oklahoma

Author: Richard C. Fox

Release date: December 7, 2009 [eBook #30620]

Most recently updated: January 5, 2021

Language: English

Other information and formats: www.gutenberg.org/ebooks/30620

Credits: Produced by Chris Curnow, Woodie4, Joseph Cooper and the Online Distributed Proofreading Team at <https://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK TWO NEW
PELYCOSAURS FROM THE LOWER PERMIAN OF OKLAHOMA ***

UNIVERSITY OF KANSAS PUBLICATIONS
MUSEUM OF NATURAL HISTORY

Volume 12, No. 6, pp. 297-307, 6 figs.

May 21, 1962

Two New Pelycosaurs from the Lower Permian of Oklahoma

BY

RICHARD C. FOX

UNIVERSITY OF KANSAS

LAWRENCE

1962

UNIVERSITY OF KANSAS PUBLICATIONS, MUSEUM OF NATURAL HISTORY

**Editors: E. Raymond Hall, Chairman, Henry S. Fitch,
Theodore H. Eaton, Jr.**

**Volume 12, No. 6, pp. 297-307, 6 figs.
Published May 21, 1962**

**UNIVERSITY OF KANSAS
Lawrence, Kansas**

**PRINTED BY
JEAN M. NEIBARGER, STATE PRINTER
TOPEKA, KANSAS
1962**

29-3001

Two New Pelycosaurs from the Lower Permian of Oklahoma

BY

RICHARD C. FOX

In the course of examining material from fissure deposits of early Permian age collected from a limestone quarry near Fort Sill, Oklahoma, the author recovered several tooth-bearing fragments of small pelycosaurs. The fragments were examined, compared with descriptions of known kinds appearing in the literature, and determined to be new genera within the Nitosauridae (Edaphosauria) and Sphenacodontidae (Sphenacodontia).

Appreciation is expressed to Prof. Theodore H. Eaton, Jr., for permission to examine the collections of the University of Kansas from Fort Sill, and for the financial assistance furnished by his National Science Foundation grant (NSF-G8624). I am grateful both to Prof. Eaton and Mr. Dale L. Hoyt for their suggestions regarding this manuscript. The accompanying figures have been drawn by the author.

Family NITOSAURIDAE

Delorhynchus priscus new genus and new species

(*delos*, Gr., evident; *rhynchos*, Gr., neuter, nostril; *priscus*, L., ancient. *Delorhynchus* is masculine because of the ending that it acquires when transliterated into Latin.)

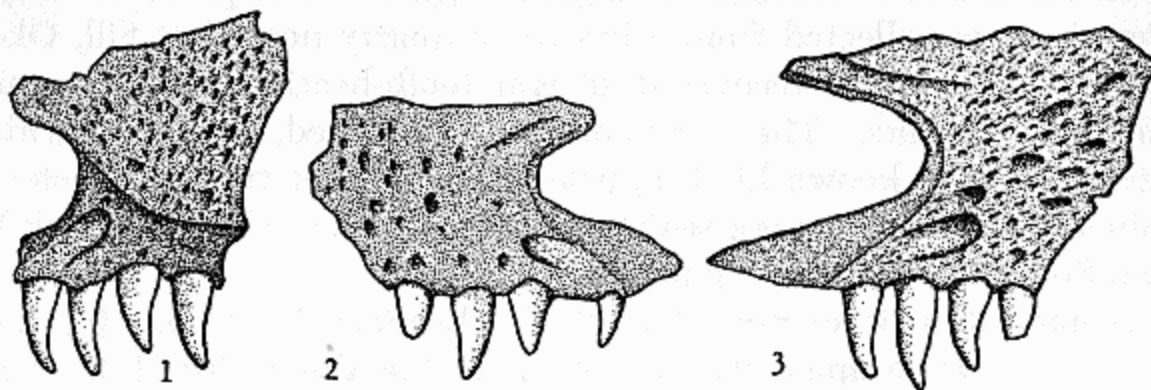
Type specimen.—Fragmentary left maxilla, bearing four teeth, KU 11117.

Referred specimens.—Fragmentary right maxilla having four teeth, KU 11118; fragmentary left maxilla having four teeth, the most posterior of which has been broken, KU 11119.

Horizon and locality.—A fissure deposit in the Arbuckle limestone at the Dolese Brothers Limestone Quarry, approximately six miles north of Fort Sill, in sec. 31, T. 4 N, R. 11 W, Comanche County, Oklahoma. These sediments are of early Permian age, possibly equivalent to the Arroyo formation, Lower Clear Fork Group of Texas (Vaughn, 1958: 981).

Diagnosis.—Small; marginal teeth conical, slender and recurved at tips; marginal tooth-row without caniniform enlargement; narial opening enlarged and bordered dorsally, posteriorly and ventrally by maxilla; maxilla with foramen opening laterally at posteroventral corner of naris.

Description (based on 3 maxillary fragments, see Table 1).—Each of the maxillary fragments bears four thecodont teeth. These are conical, slender and sharply pointed; in their distal third they are slightly recurved, laterally compressed, and have anterior and posterior non-serrated cutting edges. In medial aspect at their bases, the teeth are longitudinally striated. The bases of the teeth are circular in cross-section and are slightly bulbous. There is no caniniform enlargement of any of the teeth, the longest tooth of each fragment being differently placed in the series of teeth and little longer than the others. There is no swelling on either the internal or external surfaces of the maxillae. The teeth are in a continuous series; no diastema or maxillary step is evident.



FIGURES 1-3. *Delorhynchus priscus*, lower Permian, 6 miles north of Fort Sill, Comanche County, Oklahoma. All $\times 3$.

FIG. 1. KU 11117 (type specimen), lateral view of left maxilla.

FIG. 2. KU 11118, lateral view of right maxilla.

FIG. 3. KU 11119, lateral view of left maxilla.

The fragments have been broken along similar lines of fracture, and each is approximately rhomboidal in shape. The maxilla encircles the posterior border of the naris and extends dorsally above the naris to an extent sufficient to indicate the probable exclusion of the lacrimal bone from the narial border. At the posteroventral corner of the naris a foramen opens onto the lateral surface of the maxilla. The opening is the entrance to a canal that runs posteriorly above the tooth-row throughout the length of each specimen. Beneath the naris the maxilla extends as a broad tapering shelf, the ventral surface of which articulates with the premaxilla. The narial rim is wide, but wider ventrally than dorsally. The plane of the narial rim is oblique to the lateral surface of the maxilla. The external surface of each fragment is grooved and pitted. The ossification of each fragment appears to have been complete.

TABLE 1.—DIMENSIONS, IN MILLIMETERS, OF THREE MAXILLARY FRAGMENTS OF *DELORHYNCHUS PRISCUS*

Key:

A. Anterior height of fragment

B. Posterior height of fragment

C. Length of fragment at tooth-row

D. Dorsal length of fragment

E. Mean length of teeth
F. Anterior width of naris

CATALOGUE NUMBER AND MEAN	A.	B.	C.	D.	E.	F.
KU 11117	6.0	8.0	6.0	8.0	3.0	3.0
KU 11118	6.0	6.0	9.0	8.0	2.0	3.0
KU 11119	6.6	8.0	10.0	11.0	?	4.6
Mean	6.2	7.3	8.3	9.0	2.5	4.5

Discussion.—The Nitosauridae are small primitive edaphosaurs with a moderately elongate face, sharp subisodont teeth, little development of canines and few specializations. The jaw is of a primitive type and articulates on a level with the tooth-row. The palatal dentition is primitive (Romer, 1956:280). The nitosaurids are thought to be related to the later Caseidae, and the most obvious structural similarities are found in the postcranial skeleton (Vaughn, 1958:989). Cranial resemblances between the families are fewer, but nevertheless indicate that a nitosaurid-caseid relationship exists.

Vaughn (1958) described a small pelycosaur, *Colobomycter pholeter* (Eothyrididae, Ophiacodontia) that structurally resembles the Caseidae. This individual also was obtained from the Fort Sill locality. In Vaughn's opinion the features of *Colobomycter* indicate a close relationship between eothyridids and caseids and the possibility that the caseids may well have been of eothyridid rather than nitosaurid derivation.

In view of this historical uncertainty of the relationships between the Nitosauridae, the Eothyrididae and the Caseidae, it is well to consider how the maxillary fragments described above differ from and resemble representatives of each of these three families as reported in the literature.

Delorhynchus resembles *Colobomycter* in size. The mean extra-maxillary length of the undamaged teeth of the three fragments is 2.5 mm., equal to that reported by Vaughn (1958:985) for teeth about midway in the postcanine series of *Colobomycter*. None of the teeth of *Delorhynchus*

extends beyond the maxillary rim as far as does the canine of *Colobomycter* (3.5 mm.).

The teeth in both genera are conical and sharply pointed. The naris in each is enlarged, and the lacrimal is excluded from the narial margin in each (by inference in *Delorhynchus*.)

The differences between the maxillae of *Colobomycter* and *Delorhynchus* are most striking in the lack of canines in the latter and the correlated absence of modifications of the maxillary for support of canines. Additionally, *Delorhynchus* bears an infraorbital canal in contrast to the groove in similar position in *Colobomycter*. The recurvature of the four teeth present in the fragments of *Delorhynchus* differs from that in the teeth of *Colobomycter* in which only the canine and precanine are recurved. Vaughn implies that anterior and posterior cutting edges extend the length of the teeth in *Colobomycter*; these are restricted to the distal third of the teeth in *Delorhynchus*. The external surfaces of the maxillae of *Delorhynchus* are pitted and ridged; Vaughn was unable to discern sculpturing of the corresponding surfaces in *Colobomycter*.

Delorhynchus resembles the nitosaurids in size, the shape and sharpness of the teeth, their recurvature and the slight enlargement of their bases, the exclusion of the lacrimal bone from the narial margin (in *Mycterosaurus*) and the apparent lack of a special canine pair of teeth. Resemblances to the caseids are to be noted in the enlargement of the naris (4.5 mm. in height as opposed to 1.7 mm. in *Colobomycter*), lack of development of canines, presence of an infraorbital canal (in *Cotylorhynchus*) and absence of many replacement gaps in the marginal row of teeth.

The absence of caniniform enlargement and the extension of the maxilla dorsad of the naris exclude *Delorhynchus* from the Eothyrididae (Ophiacodontia) but are no bar to its inclusion in the Nitosauridae (Edaphosauria). The marginal teeth of *Delorhynchus* are simple and primitive, being much like those of the nitosaurids that are described in the literature.

The large narial opening and its posterior, dorsal and ventral enclosure by the maxilla, the infraorbital canal, and the sculptured external surfaces of

the maxillary fragments indicate that *Delorhynchus*, in these features at least, is close to achieving the caseid grade.

Family SPHENACODONTIDAE

Thrausmosaurus serratidens new genus and new species

(*Thrausmosaurus* is formed from the neuter Greek noun, *thrausma*, meaning fragment, and the masculine Greek noun, *sauros*, meaning reptile. The specific name, *serratidens*, is formed from the Latin *serratus*, meaning serrate, and the masculine Latin noun, *dens*, meaning tooth. The specific name is used as a substantive in apposition with the generic name.)

Type specimen.—Fragmentary left dentary, bearing five teeth, the most posterior of which is broken at the base, KU 11120.

Referred specimens.—Fragmentary ?left maxilla, having two teeth, KU 11121; fragmentary left dentary having two teeth, KU 11122.

Horizon and locality.—From the early Permian fissure deposits in the Arbuckle limestone of the Dolese Brothers Limestone Quarry, approximately 6 miles north of Fort Sill, in sec. 31, T. 4N, R. 11 W, Comanche County, Oklahoma.

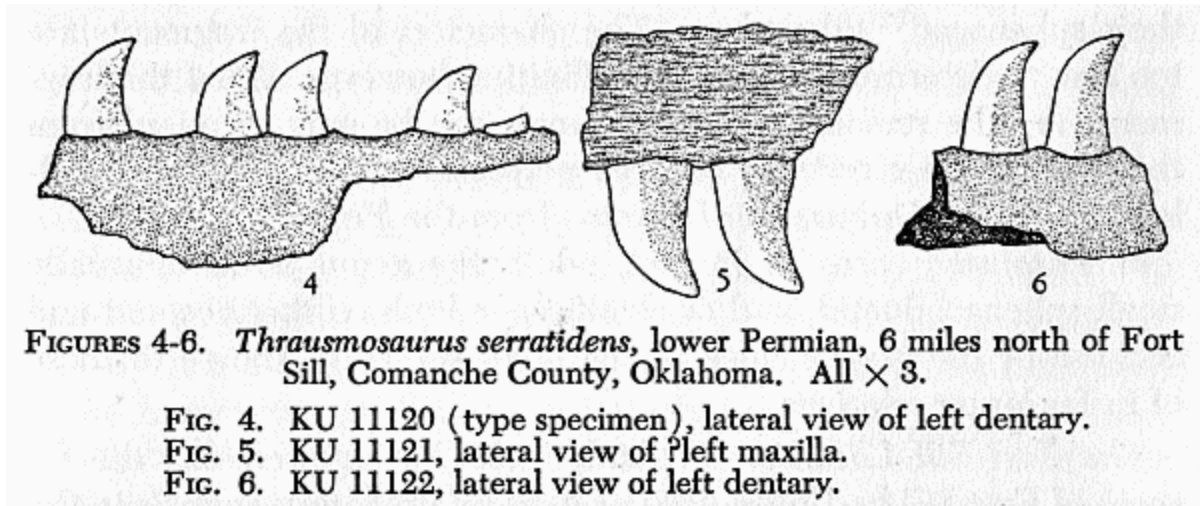
Diagnosis.—Small; teeth thecodont, compressed laterally, recurved distally, and bearing anterior and posterior cutting edges; anterior serrations limited to recurved portions of teeth, posterior serrations extending nearly entire length of teeth; lateral compression of teeth more pronounced medially than laterally; bases of teeth expanded.

Description.—The type specimen is 16 mm. long. It bears five teeth that are implanted in a straight row. Empty sockets are present between the first and second teeth, and the third and fourth teeth. The first tooth is 3.0 mm. long, the middle two are each 2.5 mm. long, and the fourth tooth is 2.0 mm. long. The fifth tooth is broken off at its base.

The empty sockets are large. The mouth of each is circular and approximately 2.0 mm. in diameter. Both sockets are 1.25 mm. deep. The bases of the teeth are expanded to fill the sockets, although the blades of the

teeth arise from only the lateral portions of the bases. The edge of the dentary rises above the bases of the teeth medially, thereby producing a small depression at the junction of each base with the dentary bone.

The lateral compression of the teeth is pronounced but asymmetrical, in that the lateral surface of each blade is more convex than the medial surface.



The recurvature of the anterior cutting edges is much more severe than that of the posterior edges, but the recurvature of both is limited to the distal half of each tooth.

The serrations of the cutting edges are not visible to the naked eye and are limited on the anterior edges of the teeth to those portions of the blades that are recurved. The posterior serrations extend nearly to the junction of the blade of each tooth with its base. The serrations tend to be more nearly crenulate than cuspidate.

A portion of the lateral wall of the dentary surrounding the Meckelian canal is present. The external surface of the wall is gently convex and smooth, without sculpturing. The internal surfaces of the canal are unmarked either by muscle scars or foramina.

The fragment is a piece from the posterior portion of the dentary, since the decrease in height from the first tooth to the fourth is pronounced.

KU 11122, a fragment of the left dentary bearing two teeth, is 7.5 mm. long. The anterior tooth is 3.0 mm. long; the posterior tooth is 3.5 mm. long. The

shape of the teeth and their implantation conform to the description of the type specimen. The lateral surface of the fragment is smooth and gently convex. What little is present of the surface bordering the Meckelian canal is unmarked.

The ?maxillary fragment bears two teeth which are 3.0 mm. long, and which conform in their characters to the type. The lateral, medial and ventral surfaces of the fragment have been sheared off, so that an exact identification of the bone is impossible. Presumably the fragment is too deep dorsoventrally to be a piece of the dentary, and no sign of the Meckelian canal is present.

Discussion.—The implantation, lateral compression, recurvature and cutting edges of the teeth borne by these fragments make clear their sphenacodontid nature. The characters of the fragments are too few to determine subfamilial affinities, however. That the fragments are the remains of adult animals can be only surmised from the lack of bones or teeth of large pelycosaurs in the extensive collections of the University of Kansas from the Fort Sill locality.

If *Thrausmosaurus* is, in fact, adult, the genus is an unusually small sphenacodontid, and of significance both on that account and because of the resemblance of the teeth presently known to those of its far larger relatives.

The Fort Sill Locality.—Peabody (1961) suggested that the fissures of Fort Sill had been used as dens by predatory animals in the early Permian, and that the unusually abundant bones in the fissures were the remains of animals eaten there by these occupants. Evidence now known to me affords an alternative explanation that is presented here as a preliminary to a more complete study of the fauna and paleoecology of these deposits currently being undertaken.

The suggestion that the skeletal material found in the fissures is the remnant of the prey of other animals is questionable because of:

1. The absence of tooth marks on the fossils.
2. The recovery from the matrix of skulls and portions of articulated skeletons that are undamaged or damaged only by pressure after burial.

3. The rarity in the deposits of animals of larger body size than *Captorhinus*, the exceptions being a few limb fragments and skull fragments of labyrinthodont or pelycosaurian nature.

4. The absence of coprolites in the matrix.

If the fissures were the dens of predators, at least some and probably many of the bones would show tooth marks. A predator feeding on other animals would be expected to leave some evidence of its habits on the bones of its prey. No such evidence is known to me, either from my own examination of several thousand bones or from the reports in the literature by others who have studied aspects of the early Permian fauna of Fort Sill.

If the predators were larger than *Captorhinus* and occupied the fissures for a long enough time to account for the accumulation of the tremendous numbers of individuals that are represented, a considerable amount of the skeletal material of the larger animals would be present in the fissure deposits. Even if for some reason the predators died in areas other than within the fissures, thereby accounting for the absence of large bones, coprolites should appear in the deposits if, in fact, the fissures were feeding places. In view of the nearly undamaged condition of many of the bones recovered from the fissures, it is reasonable to expect that fecal material would be preserved.

The character of the matrix of the deposits varies from a homogeneous clay to clay interrupted by layers of soft, limey, conglomeratic rock, to a hard, well-cemented, calcareous conglomerate. In general the bone in each kind of matrix is colored characteristically and exhibits a characteristic degree of wear. The bones entrapped in the homogeneous clay are relatively few, black, usually disarticulated, little worn and not unduly fragmented; consequently the discovery of undamaged limb bones, for example, from this kind of matrix is not unusual. The bones found in the stratified portion of the matrix are more numerous within the layers of conglomerate than between. The bones are black, brown or white, highly fragmented and waterworn to a variable degree. The fragments recovered from the hard, calcareous matrix are numerous, range in color from white through various shades of brown, to black, are highly fragmented, and are usually worn by water.

These categories for bone and matrix, however, are not mutually exclusive, since bones of any of these colors and exhibiting any degree of wear and fragmentation are found in any of the kinds of matrix described above. That water was the agent of wear is suggested by the highly polished appearance of the worn bones and pebbles that are found in the matrix.

The variability of the matrix and of the color and condition of the bones indicates that the agencies of burial and fossilization differed from time to time and that the agency of transportation of the bones from the site of burial to the fissures was running water. One can easily visualize a stream coursing the early Permian landscape that was subject to periodic flooding and droughts. Along the banks of the stream and in its pools lived a variety of microsaur, captorhinids, small labyrinthodonts and small pelycosaur. Some of the animals, after they died, were either buried near the site of their death or were swept along and buried in sediments further downstream. Burial was for a length of time sufficient to impart a color to the bones characteristic of the site in which they were buried. Later floods reexposed the sites of burial, picked up the bones and carried them to the openings into the fissures. Presumably, too, a proportion of the bones was carried to the fissures without previous burial.

The differences in wear exhibited by different bones within the same block of matrix is attributable to differences in distance that the bones were transported before final deposition. The final sites of deposition, the fissures, were inundated occasionally by floods alone, or because of changes in location of the channel of the stream at the time of flooding. The periodicity of deposition of the sediments within portions of the fissures is indicated by the stratification of the bone conglomerate mentioned earlier.

In summary, it seems that there is little or no evidence beyond the numbers of bones involved to support the hypothesis that the concentration of bones in the fissures of Fort Sill represents the remains of food of predators, and that the fissures were used as dens by their predatory occupants. On the contrary, the evidence indicates that the deposition of the bones in the fissures was secondary and that the agency of transportation, deposition and accumulation of the bones was an early Permian stream characterized by periodic flooding.

LITERATURE CITED

PEABODY, F. E.

1961. Annual growth zones in living and fossil vertebrates. Jour. Morph. 108 (1): 11-62, 69 figs., January.

ROMER, A. S.

1956. Osteology of the reptiles. The University of Chicago Press, Chicago, xxi + 772 pp., 248 figs.

ROMER, A. S., and PRICE, L. I.

1940. Review of the Pelycosauria. Geol. Soc. America, Spec. Pap., 28: x + 538 pp., 71 figs., 46 pls., 8 tables, December 6.

VAUGHN, P. P.

1958. On a new pelycosaur from the lower Permian of Oklahoma, and the origin of the family Caseidae. Jour. Paleont., 32:981-991, 1 fig., September.

Transmitted March 15, 1962.

*** END OF THE PROJECT GUTENBERG EBOOK TWO NEW
PELYCOSAURS FROM THE LOWER PERMIAN OF OKLAHOMA ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.